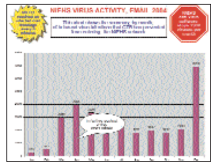
E-mail: The Good And The Bad

The proportion of e-mails that contain malware has fallen for the first six months this year compared to the same period last year. About one in 91 e-mails contained a virus or other types of malware, far less than the 1-in-35 figure of a year ago.



Designed In California, Made In China

British newspaper *The Mail* has claimed that iPods are made in Chinese factories under "slave" conditions. The paper alleged that one particular factory employed 2 lakh workers, working 15 hours a day for a monthly pay of $50 (Rs 2,300). Foxconn Electronics has sternly denied the allegations.

Enter

Konkona Sen Sharma
Actress

Although she was introduced as the daughter of actor/director Aparna Sen, Konkona made a niche for herself with her very first feature film *Mr and Mrs Iyer*. She went on to wow audiences in films such as *Page 3* and *15 Park Avenue*.

Technology to you is....

... what makes us sit up and take notice of our surroundings everyday. I mean, with every new innovation, we actually wonder "Wow! What next?"

What are your favourite gadgets?

I don't have any particular favourites especially since technology is such a double-edged sword. For example, my cell phone, a Nokia 6100, is undeniably an instrument I cannot make do without. But it can sometimes be intrusive. However, I'm curious about new innovations and do keep myself informed.

What do your online activities entail?

As an actress, work takes up much of my time, and I don't find the time for more than surfing and e-mail. But e-mailing is a key activity; I handle a lot of my work through it.

Any particular incident where technology came to your rescue?

No major life-saving incident, but technology brings small mercies to everyday living. Even with my cell phone, for example: I can actually recognise an alien number and not take the call if I don't wish to—and call back later. Isn't that a boon?

THE CORE WARS

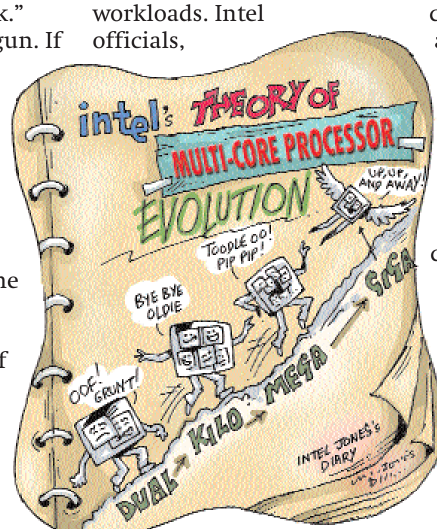
A Giga-Core Processor By 2020?

A research director at Gartner has recently said that the move from single- to dual-core processors has broken a certain barrier: "We're now in a multi-core world—there's no looking back." The core wars have begun. If you were only just getting used to the idea of dual-core, prepare to expand your imagination.

Intel officials have indicated that chips with dozens of cores might be possible by the end of the current decade. In 10 years, chips with hundreds of cores might emerge, they say. And as a natural progression, we might use the words "kilo," "mega," and "giga" to refer to the ~number of cores~ in a processor!

Intel is planning to ship quad-core chips later this year to computer manufacturers. Not to be left behind, AMD, of course, will possibly ship quad-core chips in early 2007.

It seems Intel will have the lead, in terms of both performance and energy efficiency. Speaking at a conference in mid-June in New York, Dileep Bhandarkar, architect at large for Intel in Santa Clara, said that when Intel's quad-core "Clovertown" is released in the first half of 2007, it will be a single package of two dual-core chips, and that the chips will not have the memory controller integrated into the chip. Bhandarkar did admit that integrating the memory controller directly into the chip would improve performance with some workloads. Intel officials,



Illustrations **Harsho Mohan Chattoraj**

however, think it is more important to bring a quad-core processor to the market before AMD does. Intel expects to beat AMD by a quarter or two.

According to Bhandarkar, Intel doesn't stand to lose much in the way of performance by having a multi-chip package for the quad-core processors, as opposed to AMD's design—a single piece of silicon with four cores. The Clovertown chip's performance will be helped by having independent front-side buses dedicated to each processor.

Naturally, applications that take advantage of multiple cores will need to be built, and they will. It's not clear how the chip giants are surging ahead with their plans for multi-core desktop computing in the absence of software that can use them, but that's always been the story: software follows hardware follows software. Something like Windows Vista's hardware requirements!

But Clovertown aside, is a giga-core processor conceivable? No-one knows, of course, but you can never tell with nanotechnology. What's confusing is the variety of approaches to increasing performance. What exactly will the computer of 2020 look like? Will it be based on multi-core silicon chips? Will something replace silicon? Will it be alternative computing paradigms, such as quantum and DNA computing? Will it be something exotic, like reversible computing? We live in an exponential age.

PARANOIA

Error 417: End Of World

Experts are warning that IT disasters could be a threat to the human race. The Bulletin of the Atomic Scientists (BAS)—known for its Doomsday Clock, which